

The Horn-Schunck model

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Definition 1.1

Let Ω be an compact domain in $\mathbb{R}^d$, for some $d \geq 1$. The reference and moving image are square-integrable functions that associate a gray-scale intensity to each coordinate, i.e. $I, J \in L^2(\Omega; [0, 1])$.

Definition 1.2

The deformation vector field is a coordinate transformation $\mathbf{u} : \Omega \rightarrow \Omega$. We assume that the deformation field is continuously differentiable, and that $\mathbf{u}$ and $D\mathbf{u}$ are both square-integrable, although we will in the remainder of this manuscript no concern ourselves about existence and uniqueness of solutions.

The Horn-Schunck model for measuring optical flow assumes that image intensity between images frames is conserved, leading to the following continuity equation:

$$d_t I = \partial_t I + \mathbf{u} \cdot \nabla I = 0. \quad (1.1)$$

where the time-derivative denotes the voxelwise difference between consecutive image frames, i.e. $\partial_t I = I - J$, and $\mathbf{u}$ describes the flow along which the spatial image gradient transports to compensate for image misalignment.

The Horn-Schunck model defines the optimal deformation vector field, $\mathbf{u}^*$, as the one that minimizes any residual error in Eq. (1.1), while simultaneously enforcing spatial smoothness of the deformation field. In other words,

$$\mathbf{u}^* = \arg \min_{\mathbf{u}} \left(D(\mathbf{u}) := \frac{1}{2} \int_{\Omega} (\partial_t I + \mathbf{u} \cdot \nabla I)^2 d\mathbf{x} + \frac{\alpha}{2} \int_{\Omega} \|D\mathbf{u}\|_F^2 d\mathbf{x} \right) \quad (1.2)$$

with regularization parameter $\alpha > 0$. One of the limitation of the Horn-Schunck method is that it is unable to deal with scenarios where intensity conservation between frames is violated, and Eq. (1.1) is no longer a viable assumption.

The Cornelius-Kanade model

The Cornelius-Kanade model aims to overcome this limitation by introducing an auxiliary field, $c \in L^2(\Omega; \mathbb{R})$, which captures all intensity variations between frames not associated to motion. In the context of the continuity equation, it acts as an intensity source/sink term such that

$$\partial_t I + \mathbf{u} \cdot \nabla I = c. \quad (2.1)$$

The Cornelius-Kanade model simultaneously estimates the deformation vector field and the intensity term, assuming that the intensity variation is smooth, by solving:

$$(\mathbf{u}^*, c^*) = \arg \min_{(\mathbf{u}, c)} \left(D(\mathbf{u}, c) := \frac{1}{2} \int_{\Omega} (\partial_t I + \mathbf{u} \cdot \nabla I - c)^2 d\mathbf{x} + \frac{\alpha}{2} \int_{\Omega} \|D\mathbf{u}\|_F^2 d\mathbf{x} + \frac{\beta}{2} \int_{\Omega} \|\nabla c\|^2 d\mathbf{x} \right). \quad (2.2)$$

where $\beta > 0$ is another regularization parameter for the smoothness of c .

Definition 2.1

A test field ψ is a smooth deformation vector field with compact support in Ω and satisfies $D\psi = 0$ on the boundary $\partial\Omega$. Similarly, a test function φ has compact support in Ω and satisfies $\nabla\varphi = \mathbf{0}$ on $\partial\Omega$.

Lemma 2.1 Integration-by-parts identities

Let $\mathbf{u}$ be a deformation vector field, ψ a test field, c an intensity function and φ a test function. Then:

$$\int_{\Omega} D\mathbf{u} : D\psi d\mathbf{x} = - \int_{\Omega} \Delta\mathbf{u} \cdot \psi d\mathbf{x} \text{ and } \int_{\Omega} \nabla c \cdot \nabla\varphi d\mathbf{x} = - \int_{\Omega} (\Delta c)\varphi d\mathbf{x}. \quad (2.3)$$

where $A : B$ denotes the Frobenius inner-product between matrices. Boundary terms vanish due to boundary conditions of the test function/vector field.

To find the solution to the optimization problem, we derive the Euler-Lagrange equations. These equations are derived by taking the functional derivative of the cost function D with respect to a test

field ψ and test function φ . This gives:

$$0 = \frac{d}{dt} D(\mathbf{u} + t\psi, c) \Big|_{t=0} = \left\langle (\partial_t I + \mathbf{u} \cdot \nabla I - c) \nabla I - \alpha \Delta \mathbf{u}, \psi \right\rangle_{L^2(\Omega)} \quad (2.4a)$$

$$0 = \frac{d}{dt} D(\mathbf{u}, c + t\varphi) \Big|_{t=0} = \left\langle -(\partial_t I + \mathbf{u} \cdot \nabla I - c) - \beta \Delta c, \varphi \right\rangle_{L^2(\Omega)} \quad (2.4b)$$

making use of the lemma. Via the fundamental lemma of the calculus of variations, we obtain the following list of equations (the so-called Euler-Lagrange equations):

$$(\partial_t I + \mathbf{u} \cdot \nabla I - c) \nabla I - \alpha \Delta \mathbf{u} = 0 \quad (2.5a)$$

$$-(\partial_t I + \mathbf{u} \cdot \nabla I - c) - \beta \Delta c = 0. \quad (2.5b)$$

A solution to this set of differential equations solves the minimization problem Eq. 2.2

To solve this collection of coupled partial differential equations, we typically employ numerical methods. Finite difference approximation fits nicely for image processing applications, since images are rasterized.

Definition 2.2

The image domain is discretized as

$$\Omega = \underbrace{\{1, \dots, N_1\} \times \dots \times \{1, \dots, N_d\}}_{d \text{ terms}}. \quad (2.6)$$

The reference and moving image, as well as the deformation vector field, are sampled on this rectangular grid. To denote a quantity g sampled on the image domain, we use the notation

$$g_{i_1 i_2 \dots i_d} = g(i_1, i_2, \dots, i_d), \text{ where } (i_1, i_2, \dots, i_d) \in \Omega. \quad (2.7)$$

Definition 2.3

The second-order, central, finite difference approximation of the Laplacian of a quantity g on a grid point $(i_1, i_2, \dots, i_d)$ is given by

$$\begin{aligned} (\Delta g)_{i_1 i_2 \dots i_d} &= g_{i_1+1 i_2 \dots i_d} - 2g_{i_1 i_2 \dots i_d} + g_{i_1-1 i_2 \dots i_d} \\ &\quad + g_{i_1+1 i_2 \dots i_d} - 2g_{i_1 i_2 \dots i_d} + g_{i_1 i_2-1 \dots i_d} \\ &\quad + \dots \\ &\quad + g_{i_1 i_2 \dots i_d+1} - 2g_{i_1 i_2 \dots i_d} + g_{i_1 i_2 \dots i_d-1} \end{aligned} \quad (2.8)$$

$$= 2d(\bar{g}_{i_1 i_2 \dots i_d} - g_{i_1 i_2 \dots i_d}). \quad (2.9)$$

where

$$\bar{g}_{i_1 i_2 \dots i_d} = \frac{1}{2d} (g_{i_1+1 i_2 \dots i_d} + g_{i_1-1 i_2 \dots i_d} + \dots + g_{i_1 i_2 \dots i_d+1} + g_{i_1 i_2 \dots i_d-1}) \quad (2.10)$$

Notice that $\bar{g}_{i_1 i_2 \dots i_d}$ is independent of $g_{i_1 i_2 \dots i_d}$.

Additional information on the discretization of the image gradients is given in detail in the work of Meinhardt-Lopis et al.

Lemma 2.2 The dyadic product

Give the vectors $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^d$, the dyadic product between $\mathbf{u}$ and $\mathbf{v}$ is defined as a $d \times d$ -matrix whose elements are given by $[\mathbf{u}\mathbf{v}^\top]_{ij} = u_i v_j$. This gives the following identity

$$(\mathbf{u} \cdot \mathbf{v})\mathbf{w} = (\mathbf{w}\mathbf{u}^\top)\mathbf{v}. \quad (2.11)$$

Using the discretization of the Laplacian and the dyadic product allows us to rewrite the Euler-Lagrange equations in the following form:

$$(2d\alpha + \nabla I(\nabla I)^\top)\mathbf{u} - c\nabla I = 2d\alpha\bar{\mathbf{u}} - \partial_t I \nabla I \quad (2.12a)$$

$$-\mathbf{u} \cdot \nabla I + (1 + 2d\beta)c = \partial_t I + 2d\beta\bar{c} \quad (2.12b)$$

*** Example 2.0.1 (In two dimensions: $d = 2$).** The system of equation can be represented in matrix–vector notation. In two dimensions, having $\mathbf{u} = (u, v)^\top$, this system is:

$$\begin{pmatrix} 2d\alpha + (\partial_x I)^2 & \partial_x I \partial_y I & -\partial_x I \\ \partial_x I \partial_y I & 2d\alpha + (\partial_y I)^2 & -\partial_y I \\ -\partial_x I & -\partial_y I & 2d\beta + 1 \end{pmatrix} \begin{pmatrix} u \\ v \\ c \end{pmatrix} = -\partial_t I \begin{pmatrix} \partial_x I \\ \partial_y I \\ -1 \end{pmatrix} + 2d \begin{pmatrix} \alpha & & \\ & \alpha & \\ & & \beta \end{pmatrix} \begin{pmatrix} \bar{u} \\ \bar{v} \\ \bar{c} \end{pmatrix} \quad (2.13)$$

Observe that the matrix on the left-hand side has a particular form. It can be decomposed as:

$$\begin{pmatrix} 2d\alpha + (\partial_x I)^2 & \partial_x I \partial_y I & -\partial_x I \\ \partial_x I \partial_y I & 2d\alpha + (\partial_y I)^2 & -\partial_y I \\ -\partial_x I & -\partial_y I & 2d\beta + 1 \end{pmatrix} = 2d \begin{pmatrix} \alpha & & \\ & \alpha & \\ & & \beta \end{pmatrix} + \begin{pmatrix} (\partial_x I)^2 & \partial_x I \partial_y I & -\partial_x I \\ \partial_x I \partial_y I & (\partial_y I)^2 & -\partial_y I \\ -\partial_x I & -\partial_y I & 1 \end{pmatrix} \quad (2.14)$$

If we define $\mathbf{s} = (\partial_x I, \partial_y I, -1)^\top$, $\mathbf{z} = (u, v, c)^\top$ and $J = 2d \text{diag}(\alpha, \alpha, \beta)$, then the discrete Euler-Lagrange equations can be summarized in one matrix–vector equation:

$$(J + \mathbf{s}\mathbf{s}^\top)\mathbf{z} = -\partial_t I \mathbf{s} + J\bar{\mathbf{z}} \quad (2.15)$$

Although the example is for $d = 2$, the results generalizes to higher dimensions:

Definition 2.4

Defining $\mathbf{s} = (\nabla I, -1)^\top$, $\mathbf{z} = (\mathbf{u}, c)^\top$ and $J = 2d \text{diag}(\alpha, \dots, \alpha, \beta)$, the discretized Euler-

Lagrange equations are given by:

$$(J + \mathbf{s}\mathbf{s}^\top)\mathbf{z} = -\partial_t I \mathbf{s} + J\bar{\mathbf{z}} \quad (2.16)$$

The deformation vector field and intensity term are encoded in the column vector $\mathbf{z}$. If we can invert the matrix $J + \mathbf{s}\mathbf{s}^\top$, we can express these in terms of presumably known quantities.

Lemma 2.3 The Sherman-Morrison formula

If J is an invertible, square $d \times d$ -matrix, and if $\mathbf{u}, \mathbf{v} \in \mathbb{R}^d$ are column vectors, then $J + \mathbf{u}\mathbf{v}^\top$ is invertible, iff $1 + \mathbf{v}^\top J^{-1} \mathbf{u} \neq 0$. If so, the inverse is given by

$$(J + \mathbf{u}\mathbf{v}^\top)^{-1} = J^{-1} - \frac{J^{-1} \mathbf{u}\mathbf{v}^\top J^{-1}}{1 + \mathbf{v}^\top J^{-1} \mathbf{u}} \quad (2.17)$$

Applying the Sherman-Morrison formula to Eq. 2.15 gives us:

$$\mathbf{z} = \bar{\mathbf{z}} - \frac{\partial_t I + \mathbf{s}^\top \bar{\mathbf{z}}}{1 + \mathbf{s}^\top J^{-1} \mathbf{s}} J^{-1} \mathbf{s} \quad (2.18)$$

In terms of the deformation vector field and intensity term, this is:

$$\begin{pmatrix} \mathbf{u} \\ c \end{pmatrix} = \begin{pmatrix} \bar{\mathbf{u}} \\ \bar{c} \end{pmatrix} - \frac{\partial_t I + \bar{\mathbf{u}} \cdot \nabla I - \bar{c}}{2d + \alpha^{-1} \|\nabla I\|^2 + \beta^{-1}} \begin{pmatrix} \alpha^{-1} \nabla I \\ -\beta^{-1} \end{pmatrix} \quad (2.19)$$

The Cornelius–Kanade model is to then apply this formula iteratively to obtain the minimizer and solve Eq. (2.2)